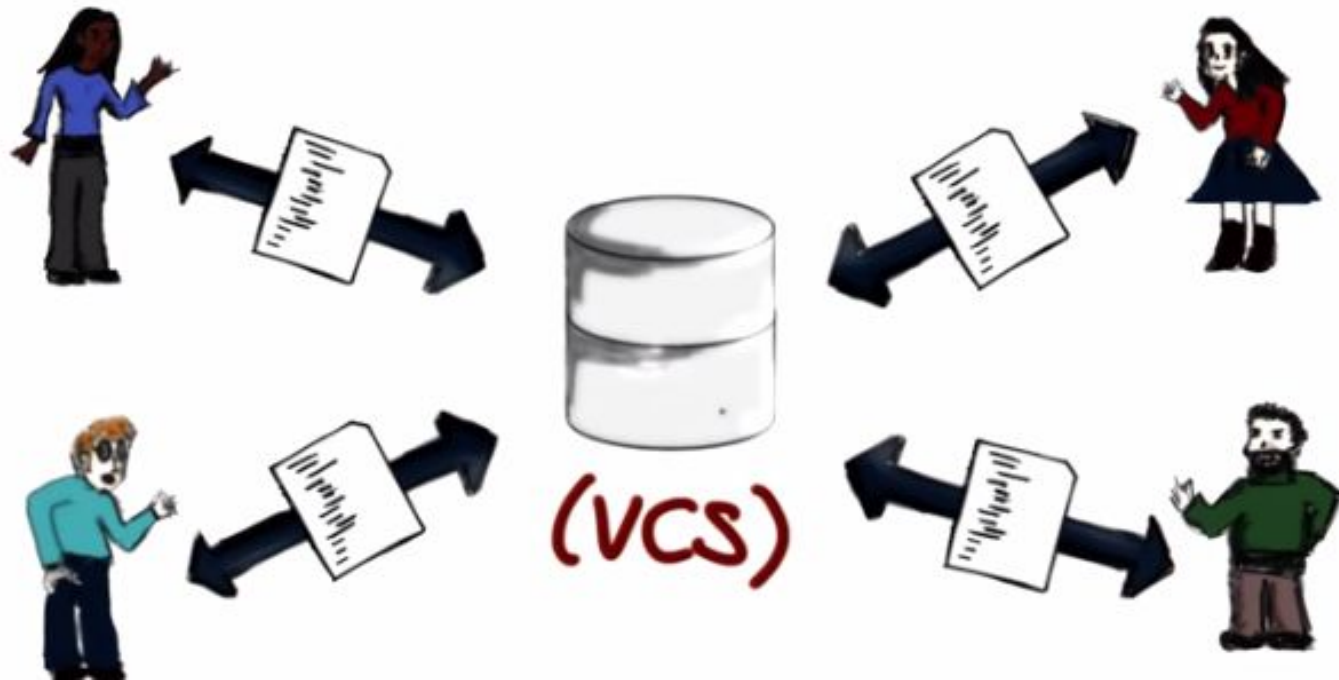


VERSION CONTROL

Lab



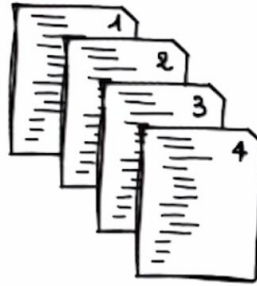
VERSION CONTROL



WHY VERSION CONTROL SYSTEMS?



Enforce discipline



Archive versions



Maintain historical information



Enable collaboration



Recover from accidental deletions or edits



Conserve disk space



ESSENTIAL ACTIONS



Add



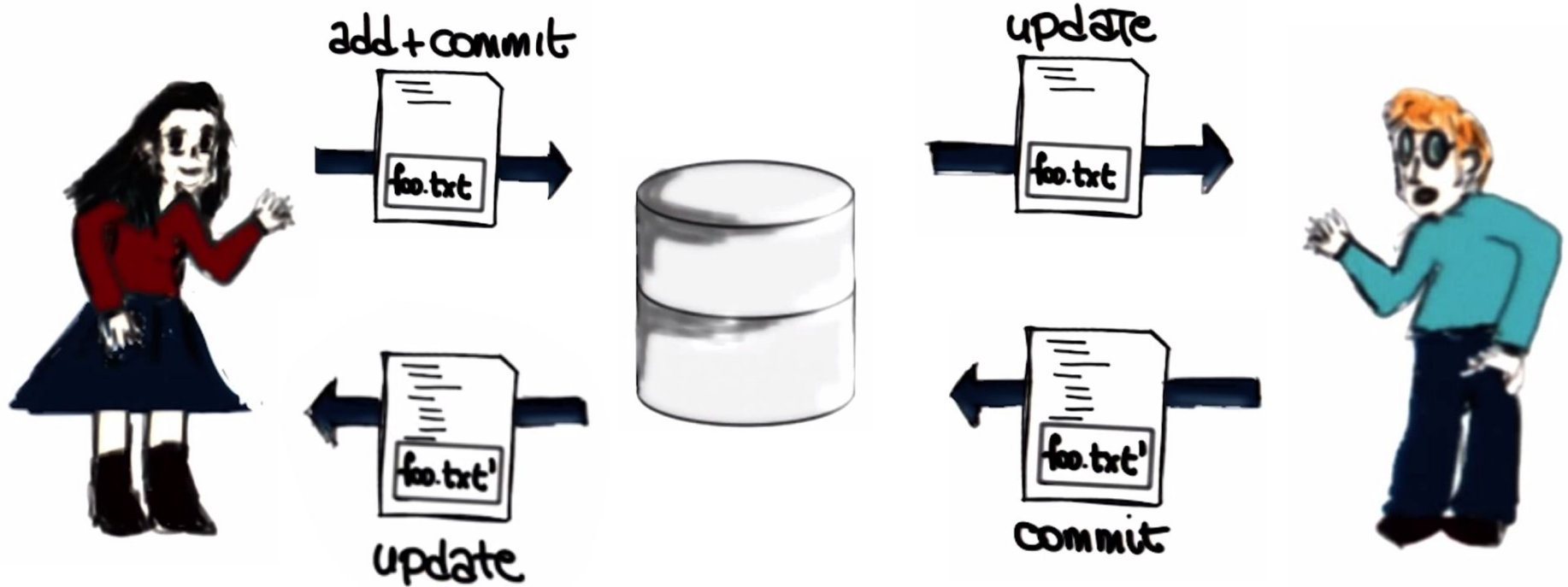
Commit



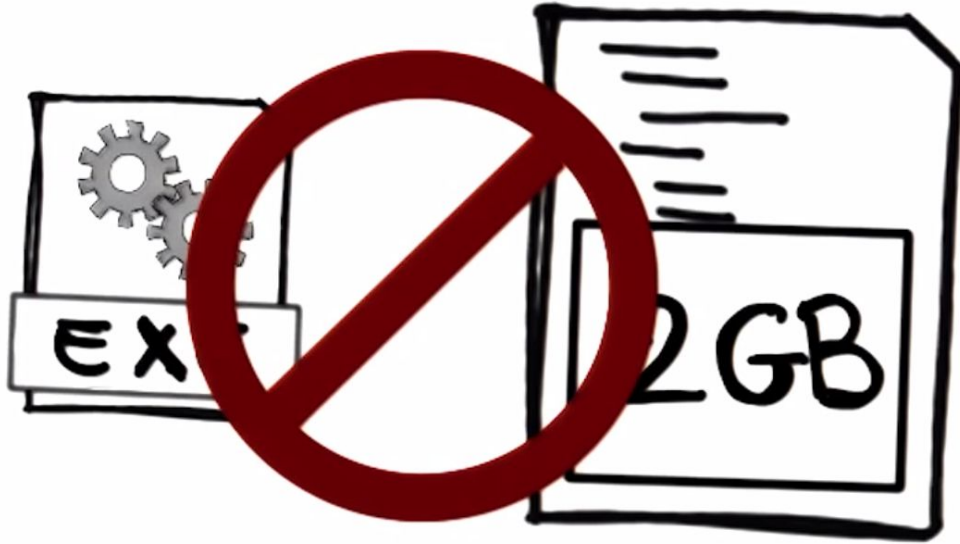
update



VCS WORKFLOW



DON'T DO IT

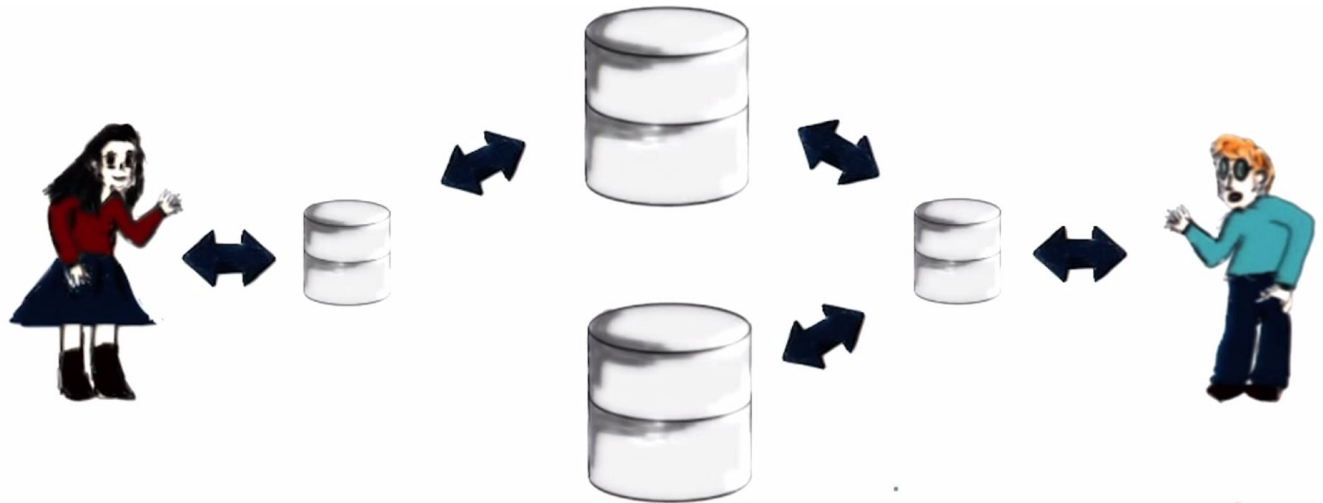


TYPES OF VCS

Centralized



Decentralized



GIT

Lab



ABOUT GIT

- DECENTRALIZED VCS
- LINUS THORVALDS
- LARGE PROJECTS IN AN EFFICIENT WAY
- FOR EXAMPLE LINUX

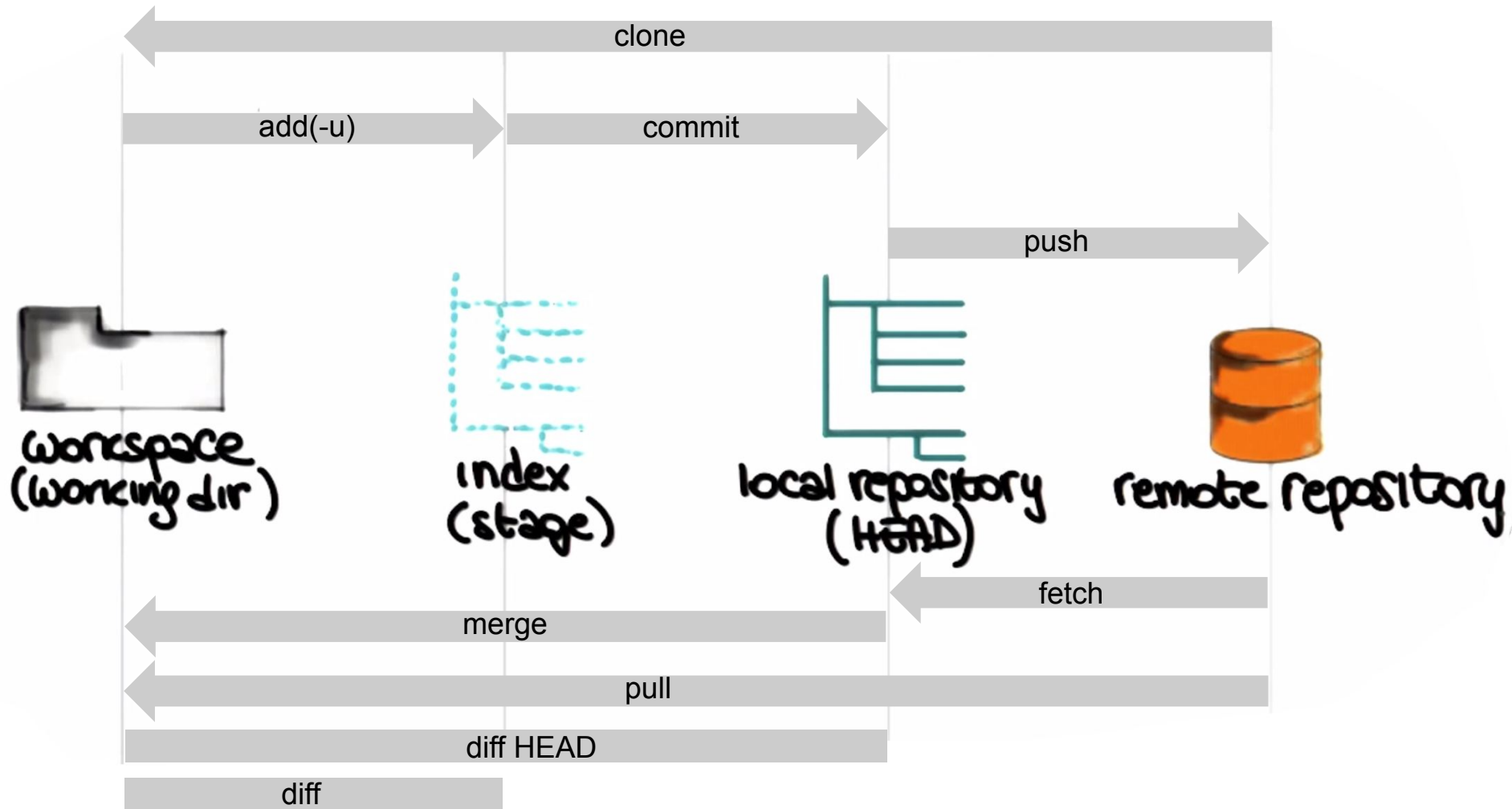


INSTALL GIT

- **LINUX: PACKAGE**
- **MAC OS: XCODE, PACKAGE**
- **WINDOWS: PACKAGE WITH INSTALLER**
- **SEE [HTTP://GIT-SCM.COM/DOWNLOADS](http://git-scm.com/downloads)**



GIT WORKFLOW



LOCAL REPOSITORIES

Create

MK

Cd

g'

A possible workflow
(This is just an example!)

Me

- some editing

g'

- git status

99

to see what files you changed

91

- `git diff [files]`

8

to see the changes

Ins

- git commit -a [-m <message>]

git

U

git

status

git

diff

git

```
show(last commit)
```



REMOTE REPOSITORIES

Copy repository

- git clone <repository>
 - repository = URL (file, http, ssh, ...)
 - creates a complete local copy of the repository
 - links it to the remote repository (origin)
(you can also link to <repository> later)

Receive changes

- git pull

Send changes

- git push



GIT HUB



LAB



PROJECTS

JHotDraw



LAB TODOS

- Clone remote repository from [GitHub](#)
- Get familiar with Git commands
 - see [this slide](#) for an overview
- Commit your initial project code using the Git commands
add, commit, pull, push...
- Input change request as an Issue at [GitHub](#)

